PIERRE-FRANCOIS ROUX

E P I G E N E T I C I S T & I N S I L I C O B I O L O G I S T

INFO

Address

IRCM - Inserm U1194

208 avenue des Apothicaires

34298 Montpellier cedex 5

France

Phone

06 67 79 62 38

Email

pierre-francois.roux@inserm.fr

SOCIAL

Website

www.pierre-francois-roux.com

ORCID

0000-0002-7047-2437

GitHub

PFRoux

LinkedIn

Pierre-Francois Roux

ResearchGate

Pierre-Francois Roux

Publons

Pierre-Francois Roux

REFERENCE

Dr Laurent Le Cam
IRCM - Inserm U1194
04 11 28 31 43
laurent.lecam@inserm.fr

Pr Anne Dejean
Institut Pasteur
01 45 68 88 86
adejean@pasteur.fr

Pr Sandrine Lagarrigue
Institut Agro Rennes-Angers
02 23 48 59 59
slagarrigue@agrocampus-ouest.fr

WORK EXPERIENCE

IRCM - Inserm U1194
Molecular Oncogenesis
PI: Dr Laurent Le Cam
Montpellier, France
2021-Present

Postdoctoral researcher
p53-dependent epigenetic and metabolic reprogramming during senescence and aging
Study of E4F1-mediated acetyl-CoA synthesis, chromatin remodeling and senescence programs using ATAC-seq, CUT&RUN, RNA-seq, metabolomics, cellular models and in vivo aging models.

Johnson & Johnson
Upstream Skin Research & Innovation
PI: Dr G. Stamatas & Dr T. Oddos
Val-de-Reuil, France
2019-2021

Postdoctoral researcher
Integrative biology for skin research and innovation
Pre-clinical and clinical research on skin biology based on multi-omics cohorts integrating microbiome, metabolome, transcriptome and phenome data.

Institut Pasteur
Nuclear Organization and Oncogenesis
PI: Dr Anne Dejean & Dr Oliver Bischof
Paris, France
2015-2019

Postdoctoral researcher
Temporal profiling of transcriptome and epigenome organization during cellular senescence
Multi-scale study of senescence in human fibroblasts using ATAC-seq, ChIP-seq, transcriptomics, metabolomics, proteomics and network inference.

INRAe
Genetics and Genomics
PI: Pr S. Lagarrigue & Dr O. Demeure
Rennes, France
2011-2014

PhD
Genome-scale SNP detection for complex traits and editome characterization
Integrative analysis of RNA-seq and genome resequencing data for QTL fine mapping, cis-eQTL analysis, allele-specific expression and RNA editing discovery.

IRSET
Cellular Signaling & Response
PI: Dr Corine Martin-Chouly
Rennes, France
2010

MEng trainee
Lipid metabolism in macrophages during inflammation in cystic fibrosis
Characterization of lipidome and cytokine responses in primary macrophages during inflammatory and infectious contexts.

UQAM
BioMed
PI: Pr Catherine Mounier & Dr Daniel Mauvoisin
Montreal, Canada
2009

MSc trainee
Role of Stearoyl-CoA Desaturase 1 in adipocyte maturation
Study of the effect of a stable knock-down of SCD1 in murine pre-adipocytes.

EDUCATION

Agrocampus Ouest & Universite Rennes 1
2009-2011

MEng in agricultural engineering
MSc in cell and molecular biology

Universite Lille 1
2006-2009

MSc in marine biology and biological oceanography
BSc in biology and ecology

Lycee Albert Chatelet
2004-2006

French CPGE BCPST-Veto

BioCampus Montpellier
2022

Animal experimentation training designer level

SKILLS - TOOLS - LANGUAGES

Epigenomics · Genomics · Genetics · Aging · Cellular senescence · Systems biology · Cancer biology · Skin biology · NGS

R	●●●●●●●●●○
Biostatistics	●●●●●●●○○○
Machine learning	●●●●●○○○○○
UNIX	●●●●●●○○○○
NGS	●●●●●●●●●○
In vivo	●●●●○○○○○○
Bench work	●●●●●●●○○○

French
NATIVE

English
FLUENT

PIERRE-FRANCOIS ROUX

E P I G E N E T I C I S T & I N S I L I C O B I O L O G I S T

INFO

Address

IRCM - Inserm U1194

208 avenue des Apothicaires

34298 Montpellier cedex 5

France

Phone

06 67 79 62 38

Email

pierre-francois.roux@inserm.fr

SOCIAL

Website

www.pierre-francois-roux.com

ORCID

0000-0002-7047-2437

GitHub

PFRoux

LinkedIn

Pierre-Francois Roux

ResearchGate

Pierre-Francois Roux

Publons

Pierre-Francois Roux

REFERENCE

Dr Laurent Le Cam
IRCM - Inserm U1194
04 11 28 31 43
laurent.lecam@inserm.fr

Pr Anne Dejean
Institut Pasteur
01 45 68 88 86
adejean@pasteur.fr

Pr Sandrine Lagarrigue
Institut Agro Rennes-Angers
02 23 48 59 59
slagarrigue@agrocampus-ouest.fr

PUBLICATION RECORD

24 original articles
Including 7 first or co-first author publications and 1 last/corresponding author publication.

Reviews and outreach
2 review articles, 1 didactic article and 1 radio communication.

Valorization
4 patents in skin biology, microbiome/metabolome biomarkers and non-invasive molecular profiling.

Articles

2026

Beyond chromatin accessibility: bulk ATAC-seq as an integrative assay to portray genomes and epigenomes. Toumi I., Kham C., Stuani L., Le Cam L., **Roux P.F.**, NAR Genomics and Bioinformatics.

2025

E4F1 coordinates pyruvate metabolism and the activity of the elongator complex to ensure translation fidelity during brain development. Di Michele M., Attina A., **Roux P.F.** et al., Nature Communications.

2023

Skin surface biomarkers are associated with future development of atopic dermatitis in children with family history of allergic disease. Sata T. et al., Skin Research & Technology.

Skin maturation from birth to 10 years of age: structure, function, composition and microbiome. Stamatas G.N., **Roux P.F.** et al., Experimental Dermatology.

Escape from oncogene-induced senescence is controlled by POU2F2 and memorized by chromatin scars. Martinez-Zamudio R.Y. et al., Cell Genomics.

2022

3-Deazaadenosine alleviates senescence to promote cellular fitness and cell therapy efficiency in mice. Guerrero A. et al., Nature Aging.

2021

An integrative multi-omic analysis reveals a major metabolic rewiring between baby foreskin keratinocytes and adult female abdominal keratinocytes. Mangez C., **Roux P.F.** et al., Experimental Dermatology.

Deciphering the role of skin surface microbiome on skin health: an integrative multi-omics approach reveals three microbial/metabolic clusters. **Roux P.F.**, Oddos T., Stamatas G., Journal of Investigative Dermatology.

Unlocking the mechanisms of cutaneous adverse drug reactions. Ondet T., **Roux P.F.** et al., JID Innovations.

2020

An unanticipated tumor-suppressive role of the SUMO pathway in the intestine unveiled by Ubc9 haploinsufficiency. Lopez I. et al., Oncogene.

AP-1 imprints a reversible transcriptional programme of senescent cells. Martinez-Zamudio R., **Roux P.F.** et al., Nature Cell Biology.

2018

Necroptosis microenvironment directs lineage commitment in liver cancer. Seehawer M., Heinzmann F., **Roux P.F.** et al., Nature.

Multi-tissue transcriptomic study reveals the main role of liver in the chicken adaptive response to dietary energy source. Desert C., **Roux P.F.** et al., BMC Genomics.

2017

Genome-wide characterization of RNA editing in chicken embryos reveals common features among vertebrates. Fresard L., **Roux P.F.** et al., PLoS One.

Snapshot: cellular senescence in pathophysiology. Martinez-Zamudio R., Robinson L., **Roux P.F.**, Bischof O., Cell.

Snapshot: cellular senescence pathways. Martinez-Zamudio R., Robinson L., **Roux P.F.**, Bischof O., Cell.

Human platelet lysate versus fetal calf serum. Fernandez-Rebollo E., **Roux P.F.** et al., Scientific Reports.

Histone variant H2A.J accumulates in senescent cells and promotes inflammatory gene expression. Contrepois K., **Roux P.F.** et al., Nature Communications.

2016

Transcriptomes of whole blood and PBMC in chicken. Desert C., Merlot E., Zerjal T., **Roux P.F.** et al., Comparative Biochemistry and Physiology.

2015

Third report on chicken genes and chromosomes. Smith J., Burt D.W., the Avian RNAseq Consortium, Cytogenetic and Genome Research.

The extent of mRNA editing is limited in chicken liver and adipose. **Roux P.F.**, Fresard L. et al., G3.

Combined QTL and selective sweep mappings revealed PARK2 and JAG2 as candidate genes for adiposity regulation. **Roux P.F.** et al., G3.

2014

Comparison of whole-genome and capture resequencing methods for SNP and genotype callings. **Roux P.F.** et al., Animal Genetics.

Re-sequencing data for refining candidate genes and polymorphisms in QTL regions affecting adiposity in chicken. **Roux P.F.** et al., PLoS One.

2013

Analysis of allele-specific expression in mouse liver by RNA-seq. Lagarrigue S., Martin L., **Roux P.F.** et al., Genetics.

Impaired function of macrophage from cystic fibrosis patients. Simonin-Le Jeune K., **Roux P.F.** et al., PLoS One.

PIERRE-FRANCOIS ROUX

EPIGENETICIST & IN SILICO BIOLOGIST

INFO

Address

IRCM - Inserm U1194

208 avenue des Apothicaires

34298 Montpellier cedex 5

France

Phone

06 67 79 62 38

Email

pierre-francois.roux@inserm.fr

SOCIAL

Website

www.pierre-francois-roux.com

ORCID

0000-0002-7047-2437

GitHub

PFRoux

LinkedIn

Pierre-Francois Roux

ResearchGate

Pierre-Francois Roux

Publons

Pierre-Francois Roux

REFERENCE

Dr Laurent Le Cam
IRCM - Inserm U1194
04 11 28 31 43
laurent.lecam@inserm.fr

Pr Anne Dejean
Institut Pasteur
01 45 68 88 86
adejean@pasteur.fr

Pr Sandrine Lagarrigue
Institut Agro Rennes-Angers
02 23 48 59 59
slagarrigue@agrocampus-ouest.fr

COMMUNICATIONS

Talks

2021 — **Metabolic reprogramming mediated by p53 during senescence and aging.** Vieillissement Occitanie, Toulouse, France.

2020 — **Towards a systems view of the senescence cell fate.** Invited speaker, Research Center in Food Toxicology TOXALIM, Toulouse, France.

Towards a systems view of the senescence cell fate. Invited speaker, IRCM, Montpellier, France.

2019 — **Towards a systems view of the senescence cell fate.** Invited speaker, London Institute of Medical Science - Imperial College, London, UK.

2017 — **A dynamic transcription factor network controls oncogene-induced senescence.** Annual Meeting of the International Cell Senescence Association, Paris, France.

2013 — **Combining genetics mapping, selective sweeps, genome resequencing and allele-specific expression reveals candidate genes for QTL regions.** Plant and Animal Genetics, San Diego, USA.

Using NGS to characterize genetics of meat-type chicken lines. European Federation of Animal Science, Nantes, France.

Selective sweeps using hapFLK combined with genome resequencing data. European Symposium on Poultry Genetics, Venice, Italy.

Posters

2022 — **E4F1-mediated control of pyruvate dehydrogenase is essential for p53-dependent senescence.** Euro Geroscience Conference, Toulouse, France.

2016 — **Dynamic multidimensional profiling defines the oncogene-induced senescence gene expression program.** International Cell Senescence Association, Rehovot, Israel.

2015 — **Dynamic multidimensional profiling defines the oncogene-induced senescence gene expression program.** AVIESAN workshop, Paris, France.

2013 — **Analysis of allele-specific expression in mouse liver by RNA-seq.** American Society of Human Genetics, Boston, USA.

Identification of candidate polymorphism in a QTL region by combining eQTL mapping with NGS data. EAAP, Nantes, France.

2011 — **Lipid accumulation in primary human macrophages from patients with cystic fibrosis.** 12e Colloque des jeunes chercheurs "Vaincre la mucoviscidose", Paris, France.

PATENTS

2022 — **Biomarkers predictive of atopic dermatitis.** JCO6146USPSP1, pending.

2021 — **Use of microbiome and metabolome clusters to evaluate skin health.** US20220142560A1, published.

Use of biomarkers to evaluate the efficacy of a composition reducing cancer therapeutic effects on skin. US20220057383A1, published.

Non-invasive method to evaluate surface skin miRNome and transcriptome. EP21306470.2.

TEACHING

ATAC-seq data analysis in practice
SIRIC Montpellier Cancer, since 2025. Practical training for cancer researchers and clinicians.

Cellular senescence and aging
Universite de Montpellier, Master 2 Biology-Health, since 2021. Molecular basis of senescence, p53 biology, metabolism and aging-related pathophysiology.

RNA-seq and epigenomic data analysis
Institut Agro Rennes-Angers / Agrocampus Ouest, since 2014. UNIX, R/Bioconductor, RNA-seq, ChIP-seq, ATAC-seq and genomic data analysis.

Cancer epigenomics
Universite Paris Diderot. Epigenomics for cancer research and clinic for MD students, MSc students and health professionals.

ALUMNI AND SUPERVISION

Former trainees

- Chaimae Kam, Master 2 Structural Biology & Genomics, 2025.
- Alizee Lanon, Master 2 Bioinformatics and Biostatistics, 2024.
- Arnaud Ferreira, Master 2 Epigenetics and Cellular Biology, 2023.
- Safiya Atiya, Master 2 AMI2B, 2023.
- Islem Toumi, Master 2 Structural Biology & Genomics, 2023.
- Pablo Garcia Idieder, Master 2 Cancer Biology, 2022.
- Lisa Salhi, Master 2 Bioinformatics, 2019.
- Camille Pere, Master 1 Cellular and Molecular Biology, 2014.
- Emile Richard, Master 2 Bioinformatics & Genomics, 2013.
- Allan Jincq, Master 1 Cellular and Molecular Biology, 2012.

Doctoral supervision and committees

- Arnaud Ferreira, PhD student, IRCM - Inserm U1194, 2023-2026. Impact of aging and cellular senescence on metabolic adaptability.
- Alexis Hucteau, PhD thesis committee, Centre de Recherches en Cancerologie de Toulouse, 2021-2024.